



Metagenomics based longitudinal monitoring of antibiotic resistome and microbiome in the inlets of wastewater treatment plants in an Indian megacity

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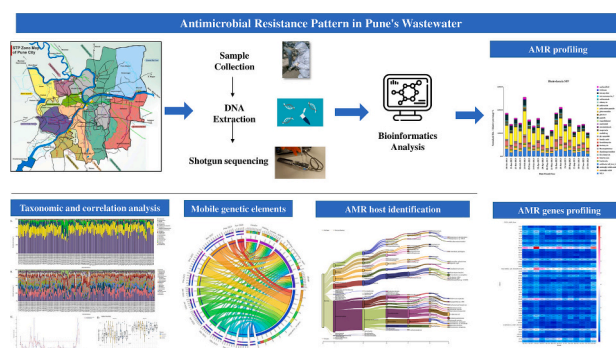
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HIGHLIGHTS

- Wastewater analysis revealed 637 ARGs tied to 29 antibiotic classes.
- SARS-CoV-2 surges linked to microbial shifts and AMR trends in Pune.
- Seasonal ARG peaks align with winter and respiratory infection treatments.
- Multidrug and beta-lactam resistance genes were most prevalent in pathogens.
- WBE proves effective for monitoring AMR and viral outbreaks.

GRAPHICAL ABSTRACT



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ABSTRACT

The growing threat of antimicrobial resistance (AMR) poses a significant global challenge, undermining advancements in healthcare, agriculture, and life expectancy. Despite its critical importance, data on population-level AMR trends, including seasonal and temporal variations, remain scarce. In this study, we conducted metagenomic analysis on 190 wastewater samples collected monthly from December 2022 to December 2023 in Pune, India, to assess the diversity, dynamics, and co-occurrence of AMR determinants. Using nanopore shotgun sequencing, we generated 87.86 Gbp of data, enabling the taxonomic classification of 157 bacterial phyla and 3291 genera. Proteobacteria dominated the microbial community, with notable seasonal shifts, including increased *Streptococcus* abundance correlating with SARS-CoV-2 viral surges in March 2023. We identified 637

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distinct antimicrobial resistance genes (ARGs) associated with 29 antibiotic classes, with multidrug, macrolide-lincosamide-streptogramin, beta-lactams, and tetracyclines genes being the most prevalent, particularly within WHO priority pathogens such as *Enterobacteriaceae* and *Pseudomonas*. Temporal normalization of ARG abundance revealed significant seasonal variability, peaking during winter, potentially driven by increased antibiotic use for respiratory infections. The integration of viral load data with AMR trends highlighted complex interactions between viral outbreaks and AMR dissemination. This study demonstrates the potential of wastewater surveillance as an early warning system for AMR, providing valuable insights into environmental and community resistance dynamics. Our results underscore the importance of integrated AMR surveillance to inform public health strategies aimed at mitigating the global AMR threat.

Abbreviations

WW	Wastewater
COVID-19	coronavirus disease 19
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
WHO	The World Health Organization
WBE	wastewater-based epidemiology
WWTP	wastewater treatment plants
NCL	National Chemical Laboratory
WWS	Wastewater-based surveillance
RRR	Replication/Recombination/Repair
IE	Integration/Excision
STD	Stability/Transfer/Defense
T	Inter-Organism Transfer
P	Phage-related biological processes
MDR	multidrug resistance
MLS	macrolide-lincosamide-streptogramin

1. Introduction

Antimicrobial resistance (AMR) is one of the most critical global health challenges of the 21st century, as it threatens to undermine the efficacy of antibiotics and other antimicrobial agents, making infections such as urinary tract infections, sepsis, and pneumonia increasingly challenging to treat (Tacconelli et al., 2018). In 2019, AMR was responsible for approximately 1.27 million deaths and contributed to an estimated 4.95 million deaths globally, highlighting the severe strain it places on healthcare systems and the economy (Murray et al., 2022). The global burden of AMR continues to escalate, and experts warn that without immediate and coordinated action, the advancements in modern medicine over the last few decades, including routine surgical procedures and cancer treatments, could be compromised by untreatable infections (Laxminarayan et al., 2013). This could result in a reversal of progress in public health, making the fight against infectious diseases more daunting than ever before.

The rise of AMR is driven largely by the misuse and overuse of antimicrobials in human medicine, agriculture, and animal husbandry (O'Neill, 2016). Antimicrobials, once life-saving drugs, have now become selective pressures in the environment that accelerate the evolution and spread of resistance genes. This spread occurs not only in pathogenic bacteria but also in commensal and environmental bacteria, creating a reservoir of resistance that can be transferred to human pathogens via horizontal gene transfer (Aarestrup and Woolhouse, 2020). Global surveillance efforts are vital to track the emergence of new resistance mechanisms, monitor resistance trends, and inform public health interventions. However, traditional AMR surveillance methods have limitations, particularly in capturing the full scope of resistance in non-clinical settings (Karkman et al., 2018).

Currently, most AMR surveillance relies on clinical isolate-based monitoring, which focuses on hospitalized patients with symptomatic

infections (Tacconelli et al., 2018). While this approach is essential for understanding resistance in healthcare settings, it fails to capture asymptomatic carriers or community-level resistance in healthy individuals, as well as AMR in environmental reservoirs. Moreover, clinical surveillance is biased toward a few priority pathogens, often underestimating the diversity of resistance mechanisms present in the broader bacterial community (Aarestrup and Woolhouse, 2020). Given that AMR evolves and disseminates not only in clinical environments but also in natural and urban ecosystems, there is a growing need for more comprehensive surveillance approaches that encompass both human and environmental factors.

Wastewater-based surveillance (WWS) offers a promising complementary strategy to traditional clinical surveillance, providing a population-wide view of AMR. Municipal wastewater serves as an aggregate sample of the microbial community from a diverse population, encompassing individuals at different stages of infection, including those who are asymptomatic, pre-symptomatic, or post-symptomatic (Ng et al., 2017; Diemert and Yan, 2019). This approach is particularly advantageous because it allows the detection of AMR without the need for individual sampling or ethical concerns regarding privacy (Auguet et al., 2021). Furthermore, wastewater contains bacteria and genetic material from hospitals, households, and other community sources, providing a comprehensive snapshot of resistance at the population level (Pärnänen et al., 2019). Studies have demonstrated that WWS can effectively monitor the prevalence of antibiotic-resistant bacteria (ARB) and resistance genes (ARGs), and may even serve as an early warning system for emerging resistance threats (Hutinel et al., 2019; Flach et al., 2021; Javvadi and Mohan, 2024).

Metagenomic sequencing, combined with WWS, represents a cutting-edge approach for AMR surveillance. Unlike culture-based methods, metagenomics allows for the direct detection of ARGs in environmental samples, bypassing the need for bacterial isolation and enabling the identification of resistance mechanisms across a broader range of bacteria, including those that are difficult to culture (Hendriksen et al., 2019). This technique provides a comprehensive view of the resistome, the collection of all resistance genes within a microbial community, offering insights into the co-occurrence of resistance traits, potential reservoirs of resistance, and the dynamics of AMR transmission (Huijbers et al., 2020).

In this study, we aim to harness the potential of metagenomic sequencing of untreated wastewater to assess AMR prevalence and diversity at the community level. Specifically, we explored the abundance and distribution of ARGs across different antibiotic classes and geographic locations. By comparing seasonal variations and identifying correlations between resistance genes, this study will provide valuable insights into the population-level dynamics of AMR. Furthermore, our work will contribute to the growing body of evidence supporting the use of WWS as a viable, cost-effective tool for large-scale AMR monitoring, complementing traditional clinical surveillance methods and potentially informing public health interventions aimed at mitigating the spread of drug-resistant infections.

2. Methods

2.1. Sample collection and processing

Grab samples were collected twice in a week from the inlets of 10 STPs spanning Pune city, India, between December 2022 to December 2023 (Supplementary_File.Table.1, Fig. S1). Around 500 ml grab samples were collected in sterile plastic bottles and kept at 4 °C during transport. Upon arrival at the laboratory, biosafety protocols were strictly followed for further processing. Sample bottles were surface sterilized with 70 % ethanol and samples were pasteurized at 60 °C for 1 h. Subsequently, 200 ml water samples were transferred to 250 ml HDPE bottles and centrifuged for 10 min at 4 °C, to collect the sludge and stored at –80 °C for downstream process.

2.2. Metagenomic DNA extraction and shotgun sequencing

The total DNA was extracted from 250 mg of sludge samples using DNeasy PowerSoil Pro Kit (Qiagen) following the manufacturer's instructions. The DNA quality and concentrations were checked by Nanodrop Lite Spectrophotometer (Thermo Fisher Scientific) followed by its qubit analysis on Qubit fluorometer (Thermo Fisher Scientific) using Qubit Broad range (BR) assay kit and Qubit High sensitivity (HS) assay kit (Thermo Fisher Scientific, Q32853 and Q32851 respectively). The DNA was stored at –80 °C until further processing. Only those DNA that had a purity of 1.8 (OD 260/280) were further processed. The library preparation was performed using the sequencing kit SQK-NBD-114.96 as per the manufacturer's instructions. The resultant library was quantified by Qubit High sensitivity (HS) assay kit, and around 500 ng of the library was loaded on FLO-MIN114 R10 flowcells (Fig. S2).

2.3. Data processing and bioinformatics analysis

Raw sequencing reads were processed for base calling and demultiplexing using Guppy (version 6.5.7) (Wick et al., 2019). For microbial diversity assessment, data rarefaction was performed using the single_rarefaction.py script along with the rarefy even depth function implemented in both QIIME (Caporaso et al., 2010) and R (McMurdie and Holmes, 2013). Taxonomic classification of metagenomic reads was executed with Kaiju (version 3), which utilizes a six-frame translation of each read, allowing for accurate classification against the NCBI BLAST nr protein database (Menzel et al., 2016). Alpha diversity metrics, specifically Chao1 and Shannon indices, were calculated using the Phyloseq package in R (McMurdie and Holmes, 2013). The analysis of antimicrobial resistance (AMR) was conducted using the DeepARG tool, which integrates DIAMOND (Buchfink et al., 2014) for sequence alignment. The alignment parameters were set to –min-score 60, –id 60, and –more-sensitive mode, as recommended in prior studies (Pearson, 2013). Identification of mobile genetic elements (MGEs) was achieved through mobileOG-db (Data Version: Beatrix), categorizing MGEs into five primary mobileOG categories: Replication/Recombination/Repair (RRR), Integration/Excision (IE), Stability/Transfer/Defense (STD), Inter-Organism Transfer (T), and Phage-related biological processes (P) (Tansirichaiya et al., 2019). For identifying AMR hosts, we employed a manual extraction method followed by taxonomic classification using the Kraken tool (Wood et al., 2019). Visualization of the data was performed with the Shanky plot in the R Pavian package (Breitwieser and Salzberg, 2020). To quantify the abundance of ARGs across the samples, the number of ARG copies was assessed. For comparative analysis of ARG abundance between samples, we normalized the total ARG counts to the total number of gigabase pairs (Gbp) in each sample, calculated using the following formula:

$$\text{Normalized copy number of ARGs} = \frac{\text{Total ARG count}}{\text{Size of the dataset in Gbp}}$$

Here, the dataset size in Gbp is determined by dividing the total nucleotide count of the library by 1×10^9 , corresponding to 1 Gbp.

3. Results and discussion

A year-long longitudinal study was conducted to explore microbial diversity and assess the prevalence of antibiotic resistance (ARGs) in Pune City's wastewater system during the COVID-19 pandemic. A total of 190 wastewater samples were collected from 10 STP locations across the city, covering an estimated population of 7.1 million residents, from December 2022 to December 2023. Utilizing nanopore MinION sequencing technology, the study generated a total of 87.86 gigabases (Gbp) of sequencing data, comprising 3.64 million reads with an average quality score of 8.86 and an average read length of 5.1 kilobases.

3.1. The effect of SARS-CoV-2 wave on microbial communities

A total of 157 bacterial phyla and 3291 genera were identified across all sampled locations, reflecting a rich and diverse microbial ecosystem. The top 20 most abundant phyla and genera contributing to overall bacterial diversity are illustrated in Fig. 1A and B. *Proteobacteria* emerged as the dominant phylum, accounting for 52.92 ± 8.03 % of the total microbial community. Followed by *Bacteroidetes* (20.86 ± 5.73 %), *Firmicutes* (8.58 ± 4.23 %), *Actinobacteria* (4.61 ± 4.96 %), and *Verrucomicrobia* (0.71 ± 0.49 %) across all locations and seasons (Fig. 1A). At the genus level, the most prevalent taxa included *Arcobacter* (7.95 ± 4.73 %), *Prevotella* (4.72 ± 2.98 %), *Cloacibacterium* (3.09 ± 2.06 %), *Rivicola* (2.34 ± 1.58 %), *Acinetobacter* (2.47 ± 1.95 %), and *Pseudomonas* (1.74 ± 1.56 %) (Fig. 1B). These findings are consistent with existing literature, which identifies *Proteobacteria* and *Bacteroidetes* as the predominant phyla in wastewater microbiome reported from China, Germany, Ireland, Malaysia, Netherlands, United Kingdom, United States of America and Global consortium spread across 23 countries (Singh et al., 2022; Su et al., 2017; Xie et al., 2021; Nummerger et al., 2019; Do et al., 2019; Wu et al., 2019; Verburg et al., 2021; LaMartina et al., 2021; Azli et al., 2022; Brumfield et al., 2022). Moreover, the presence of *Arcobacter*, *Prevotella*, and *Cloacibacterium* in our samples reinforces their established role as indicators of fecal contamination in aquatic environments (Collado et al., 2008). Despite the diverse microbial community observed, bacterial diversity remained stable across different months, locations, and seasons, with no significant differences detected ($p > 0.05$), however an increase in alpha diversity was noted during the winter season (Fig. 1D). This seasonal variation could be attributed to several environmental factors, such as temperature shifts, which may favor the growth of specific bacterial taxa adapted to colder conditions, thereby enhancing microbial diversity during the winter season (Fang et al., 2023).

Interestingly in March 2023, a significant shift in the microbial composition of wastewater was observed at both the phylum and genus levels. The relative abundance of *Actinobacteria* increased markedly, accompanied by a pronounced rise in the genera *Streptococcus*, *Azonexus*, *Dechloromonas*, *Acidovorax*, and *Flavobacterium*. In contrast, previously dominant genera such as *Acinetobacter*, *Prevotella*, and *Arcobacter* showed a substantial decline during this period (Fig. 1A, B). To understand this microbial shifts, we correlated the bacterial taxonomic data with SARS-CoV-2 viral load measurements from the same wastewater samples published by our group as a separate study (Rajput et al., 2024a, 2024b, <https://www.pkc.org.in/pkc-focus-area/health/waste-water-surveillance/www-covid-dashboard-pune/>). Notably, we found that the increase in *Streptococcus* and other genera coincided with a heightened SARS-CoV-2 viral load in wastewater. The rise in *Streptococcus* is particularly notable, as it is a known respiratory and opportunistic pathogen. Its increase in the wastewater samples reflects a broader trend of weakened immune responses within the population due to concurrent viral outbreaks of SARS-CoV-2, which has been shown to contribute to increased secondary bacterial infections (Avire et al., 2021). This



Fig. 1. Bacterial community composition, diversity, and SARS-CoV-2 viral load across Pune city. 1A: Relative abundance of the top 20 bacterial phyla identified from wastewater samples. 1B: Relative abundance of the top 20 bacterial genera in the community composition. 1C: Temporal trend of SARS-CoV-2 viral load observed across Pune city from December 2022 to December 2023. 1D: Alpha diversity analysis of bacterial communities across three seasons: Winter, Summer, and Monsoon.

observation aligns with reports from the CDC and WHO, which documented a rise in severe *Streptococcus* infections, including *Streptococcus pneumoniae* and *Streptococcus pyogenes*, following the COVID-19 pandemic (Mohapatra et al., 2023; Massese et al., 2024). These infections are often seen in populations with compromised immune systems post-viral infection (<https://www.who.int/emergencies/disease-outbreak-news/item/2022-DON429>). Conversely, the decline in *Acinetobacter*, *Prevotella*, and *Arcobacter* might be attributable to increased antibiotic use during the viral surge. These genera are commonly associated with human fecal matter, and their decline could reflect selective pressures resulting from widespread antimicrobial treatments during viral outbreaks (Posada-Perlaza et al., 2019). Further, the rise of *Azo-nexus*, *Dechloromonas*, *Acidovorax*, and *Flavobacterium*, on the other hand, highlights their environmental roles, particularly in bioremediation, dechlorination, and nutrient cycling (Jangir et al., 2016; Wu et al., 2019; jmore et al., 2013; Brumfield et al., 2022). The co-occurrence of these genera with elevated viral loads points to the potential role of wastewater-based microbial surveillance not only in tracking viral outbreaks but also in monitoring secondary bacterial infections and

environmental shifts. These findings provide a critical understanding of the complex interactions between viral outbreaks and microbial communities in wastewater, underscoring the value of integrating microbial and viral data for comprehensive public health surveillance.

3.2. Temporal dynamics of antimicrobial resistance in wastewater

The metagenomic analysis of wastewater samples revealed a diverse array of antimicrobial resistance genes (ARGs), conferring resistance to 29 antibiotic classes. Multidrug resistance (MDR) genes were predominant, representing 38.21 ± 4.68 % of the total ARGs detected, followed by those associated with macrolide-lincosamide-streptogramin (MLS) antibiotics (15.34 ± 2.41 %), beta-lactams (8.49 ± 2.24 %), tetracyclines (8.49 ± 1.92 %), and aminoglycosides (5.28 ± 1.30 %) (Fig. 2A, B). The high prevalence of MDR genes aligns with global patterns, where MDR is increasingly detected in urban wastewater systems due to the widespread and often indiscriminate use of antibiotics, particularly during the COVID-19 pandemic (Mukherjee et al., 2021; Majumder et al., 2020; Kariyawasam et al., 2022). The identification of

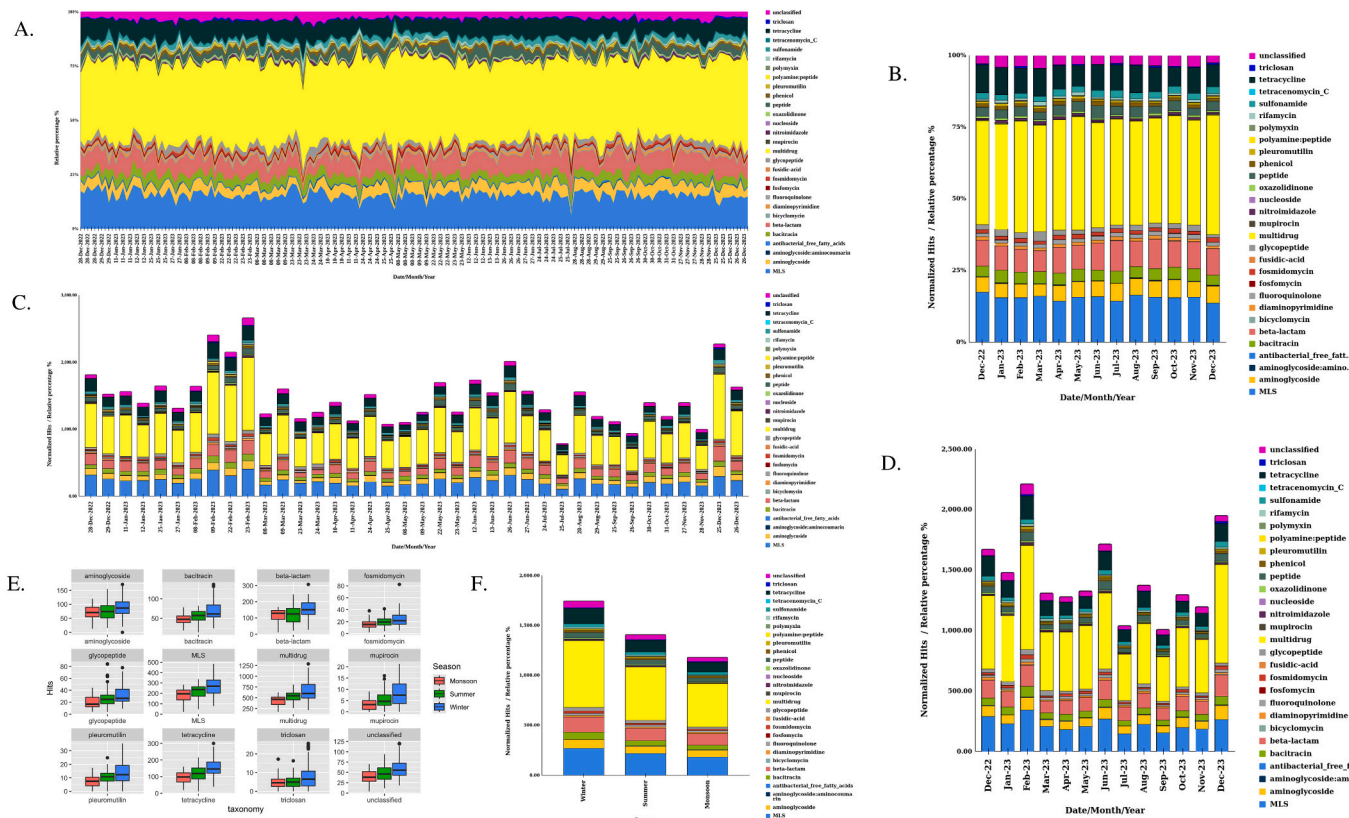


Fig. 2. Antimicrobial Resistance (AMR) profiling of community wastewater. 2A–B: Temporal trends of antimicrobial resistance (AMR) classes across sampling months. 2C–D: Normalized (per Gbp) absolute abundance of AMR classes from metagenomic sequencing data. 2E: Significant AMR classes showing variation in abundance with seasonal changes. 2F: Seasonal trends of normalized absolute abundance of AMR classes.

beta-lactam resistance is particularly concerning given the reliance on beta-lactams (e.g., penicillins, cephalosporins) as first-line treatments for many bacterial infections globally (Pandey and Cascella, 2023). Additionally, the significant presence of MLS resistance highlights the extensive use of macrolide and lincosamide antibiotics, which are widely employed in both human medicine and agriculture (Ashwini et al., 2023). These antibiotics, frequently used in clinical and agricultural settings, contribute to the enrichment of ARGs in both clinical and environmental reservoirs, a situation exacerbated by inadequate treatment of wastewater from agricultural and urban sources (Lipsitch et al., 2002; World Health Organization: WHO, 2017a, 2017b).

To ensure accurate temporal comparisons, we employed a normalization approach that assessed AMR abundance on a per-gigabase (per GB) basis, minimizing the effects of variable sequencing depths across different periods (Rolbiecki et al., 2023). Despite the relative stability in the composition of AMR classes, our results revealed substantial fluctuations in the overall AMR load over the study period. Notably, the AMR load peaked in February 2023 at 2212 hits per GB, marking a significant rise from 1668 hits per GB in December 2022, followed by a decrease to 1476 hits per GB in January 2023. This spike likely reflects increased antibiotic consumption during the winter months, often associated with a rise in respiratory infections and corresponding medical interventions (Fig. 2C) (Romaszko-Wojtowicz et al., 2024; Danion et al., 2023). The seasonal resurgence of *Streptococcus* taxa during winter may have also contributed to this elevated AMR load (Kennis et al., 2022). A gradual decline in AMR load was observed between March and April 2023, with values dropping to 1308 and 1277 hits per GB, respectively (Figs. S3, S4), likely reflecting reduced antibiotic usage following the peak infection period. However, a slight resurgence in AMR load occurred in June (1713 hits per GB) and again in December 2023 (1951 hits per GB), suggesting seasonal influences on

antibiotic prescription practices and disease prevalence. These trends are consistent with previous studies that have reported similar seasonal variations in AMR across environmental samples (Serletti et al., 2023). The reduced AMR loads during July, August, and September 2023 (1038, 1372, and 1005 hits per GB, respectively) likely correspond to a decrease in antibiotic usage during the summer months. This seasonal pattern emphasizes the complex interplay between human health practices, environmental factors, and microbial resistance dynamics (Read et al., 2024). Our findings underscore the importance of continuous monitoring of AMR in wastewater systems as an essential public health surveillance tool. By linking AMR trends with temporal and seasonal fluctuations, we can gain a deeper understanding of environmental reservoirs of resistance and inform strategies aimed at curbing the spread of AMR in both clinical and environmental contexts. The observed interplay between the COVID-19 pandemic and AMR trends further highlights the need for integrated approaches to surveillance and management practices.

3.3. Occurrence of ARGs and its host identification

Our comprehensive metagenomic analysis uncovered 637 distinct antimicrobial resistance genes (ARGs), providing critical insights into the dissemination of antimicrobial resistance (AMR) within urban wastewater systems. Among these, multidrug resistance (MDR)-associated ARGs were highly prevalent, constituting 22.5 % of the total ARGs identified, underscoring the escalating threat posed by MDR bacteria, particularly in densely populated urban settings. ARGs conferring resistance to beta-lactams (20.1 %) and macrolide-lincosamide-streptogramin (MLS) antibiotics (12.6 %) were also abundant, alongside those targeting aminoglycosides (8.2 %), tetracyclines (7.9 %), and fluoroquinolones (5.0 %) (Figs. 3A, S3E). While no prior studies have

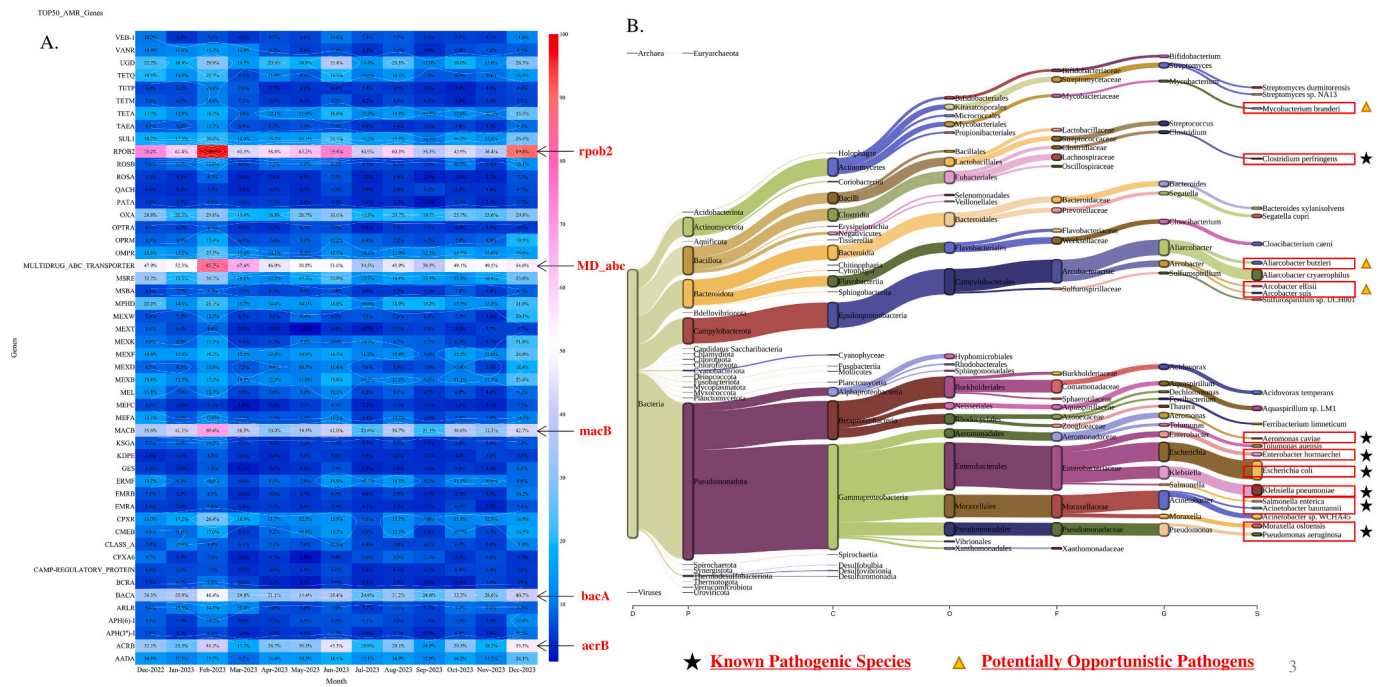


Fig. 3. AMR gene abundance and host profiling in community wastewater. 3A: Heatmap representing the top 50 most abundant AMR genes detected in wastewater samples. 3B: Graph showing AMR gene hosts, linking resistance genes to specific bacterial taxa in wastewater.

systematically monitored AMR trends in sewage treatment plants (STPs) within Pune city, comparisons with historical data from riverine ecosystems offer a valuable perspective. Marathe et al. (2017) reported 175 ARG subtypes in river samples, and Rajput et al. (2021) identified 278 ARG subtypes. The present study identified more than double the ARG diversity previously documented in river systems, indicating a marked increase in ARG abundance and diversity in urban wastewater, particularly in the post-COVID-19 era. However, the direct quantitative comparisons between river and wastewater systems have inherent limitations due to differences in environmental matrices, microbial community composition, sampling and analytical methodologies. Nevertheless, the observed increase in ARG diversity highlights a substantial enrichment of resistance determinants in urban wastewater environments, likely driven by intensified antibiotic use during the pandemic, emphasizing the need for sustained AMR surveillance in urban settings.

Prominent ARGs identified included *rpob2*, multidrug ABC transporter genes, *macB*, *bacA*, *acrB*, *oxa*, *ugd*, and *msrE*, each exceeding a relative abundance threshold of 30.5 % (Figs. S5A, S6, S7, S8). Similar patterns have been observed in riverine systems (Rajput et al., 2021), where *rpob2* and multidrug ABC transporter genes were among the most dominant. Notably, *rpob2* (6.42 %)—linked to rifampicin resistance—highlights significant exposure to this antibiotic, likely due to its widespread use in tuberculosis (TB) treatment (Bobbá and Khader, 2023). Multidrug ABC transporter genes (5.94 %), indicative of bacterial mechanisms for expelling multiple antibiotic classes, underscore the growing prevalence of MDR strains (Orelle et al., 2019). The abundance of *macB* (4.03 %), a component of the MacAB-TolC efflux pump, likely reflects the increased prescription of macrolides during the COVID-19 pandemic for respiratory infections (Sterenczak et al., 2020). Additional ARGs, such as *bacA* (3.6 %) associated with bacitracin resistance and *acrB* (2.96 %) linked to efflux-mediated resistance, highlight the contributions of both healthcare and agricultural practices to ARG enrichment in wastewater. Of particular concern was the detection of *oxa* genes (2.41 %), encoding OXA-type beta-lactamases, which confer resistance to carbapenems—critical antibiotics for treating MDR infections (Sah et al., 2019). The presence of *ugd* (2.09 %), associated with

polymyxin resistance, and *msrE* (1.87 %), linked to macrolide resistance, further underscores the necessity for ongoing surveillance, as these genes mark resistance to last-resort antibiotics.

Taxonomic classification of ARG-associated reads revealed that 99.32 % originated from bacterial taxa, spanning 41 phyla and 1574 genera (Fig. 3B). The phylum *Pseudomonadota* was predominant, representing 59.65 % of sequences, consistent with its documented association with clinically relevant ARGs, particularly in healthcare settings (Leão et al., 2022). Other notable phyla included *Bacteroidota* (10.7 %), *Bacillota* (10.59 %), and *Campylobacterota* (9.73 %), which are commonly associated with human and animal microbiota, suggesting fecal contamination as a significant driver of ARG proliferation in wastewater. At the genus level, *Escherichia* emerged as the most prevalent ARG host, comprising 8.06 % of sequences. This finding is particularly concerning given the classification of *Escherichia coli* as a WHO priority pathogen, especially strains resistant to third-generation cephalosporins and fluoroquinolones (Lin et al., 2019; Bidell et al., 2016). Other notable genera included *Acinetobacter* (7.91 %), with *Acinetobacter baumannii* recognized for its multidrug-resistant strains (Eliopoulos et al., 2008), and *Aliarcobacter* (6.64 %), implicated in human infections (Simaluiza et al., 2021). *Pseudomonas* (5.27 %) and *Klebsiella* (5.24 %), including *K. pneumoniae*, another WHO critical priority pathogen (WHO, 2017b), were also prominent contributors to the ARG profile. Additionally, genera such as *Aerobacter* (2.74 %), *Enterobacter* (2.48 %), *Aeromonas* (2.33 %), *Bacteroides* (2.27 %), *Streptomyces* (2.15 %), and *Acidovorax* (2.09 %) were identified, highlighting the extensive diversity of ARG hosts in urban wastewater. These findings underscore urban wastewater systems as reservoirs for ARGs, reflecting community and ecosystem health.

3.4. Temporal analysis of mobile genetic elements

Mobile genetic elements (MGEs) play a key role in the dissemination of antimicrobial resistance genes (ARGs), significantly contributing to the rise of multidrug resistance. MGEs enhance horizontal gene transfer (HGT), allowing the movement of genetic material between bacterial populations, which presents a considerable threat to public health

(Partridge et al., 2018). Our analysis, conducted using *mobileOG-db*, identified five major classes of MGEs in wastewater: *integration/excision*, *phage*, *replication/recombination/repair*, *stability/transfer/defense*, and *transfer*. Of these, *integration/excision* was the most prevalent, accounting for 48 % of the total MGEs, followed by *replication/recombination/repair* (21.8 %), *stability/transfer/defense* (11.27 %), *phage* (9.9 %), and *transfer* (8.9 %) (Fig. 4). Overall, the data suggests a dynamic MGE landscape with potential implications for antimicrobial resistance (AMR). Moreover, the dominance of integration/excision mechanisms, which are known to be pivotal in facilitating the incorporation and dissemination of ARGs, leading to a more robust and widespread resistance phenotype in bacterial populations (Mullany et al., 2002). The

high presence of *phage* elements suggests active phage-mediated gene transfer, a key pathway for ARG mobilization (Borodovich et al., 2022). Furthermore, the detection of *replication/recombination/repair* elements (21.8 %) indicates potential opportunities for genetic recombination, enabling the generation of novel ARG combinations and further complicating resistance patterns (Benler and Koonin, 2022). Intriguingly, temporal analysis revealed that MGE prevalence correlated strongly with peaks in AMR load, with the highest detection of MGEs coinciding with the AMR spikes observed in December 2022, February 2023, and December 2023. This temporal alignment reinforces the role of MGEs as key drivers in the escalation of ARGs during periods of heightened resistance. The elevated presence of MGEs during these peak

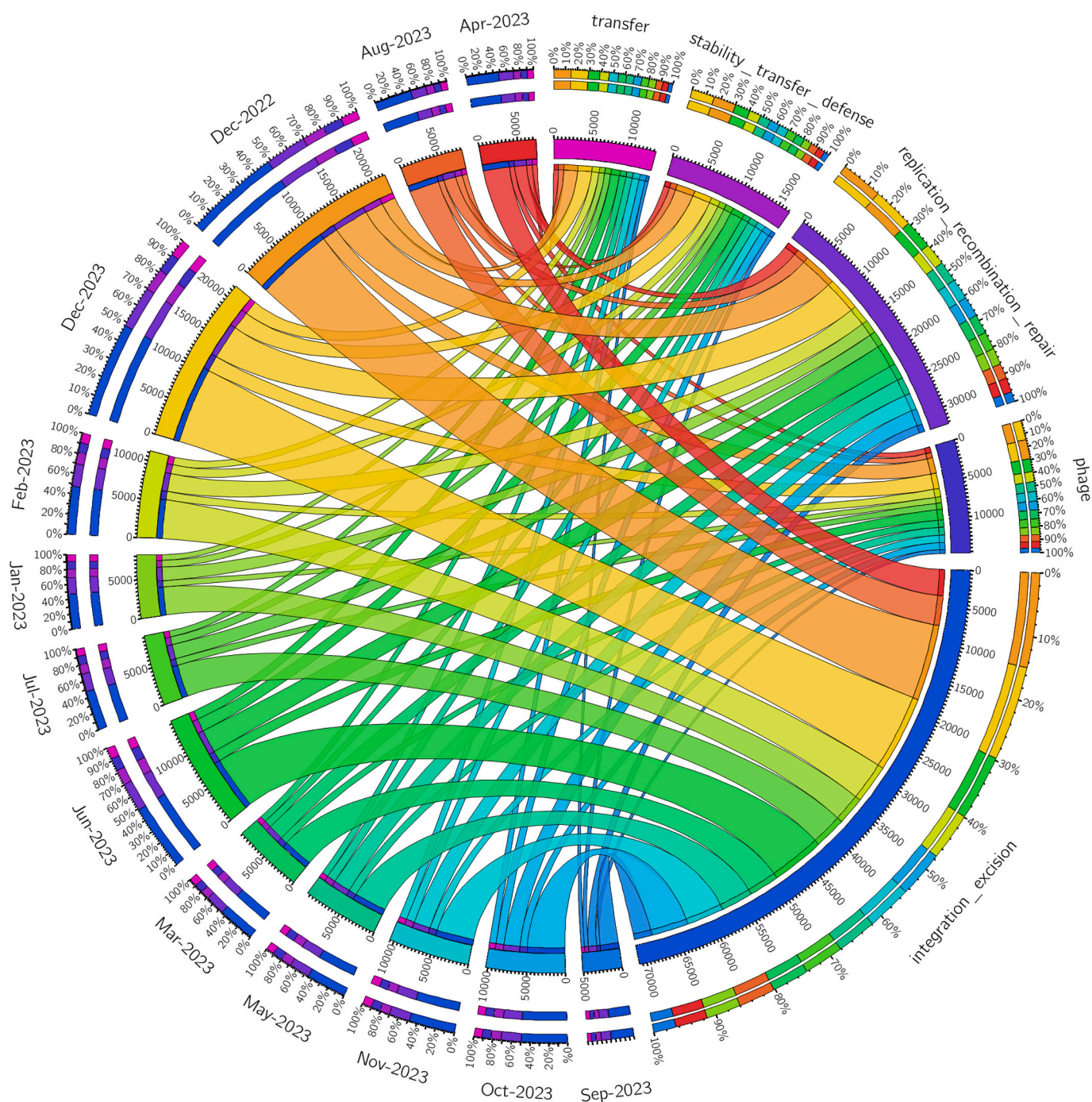


Fig. 4. Relative and absolute abundance of mobile genetic elements (MGEs) in community wastewater. Graph illustrating the relative and normalized absolute abundance of MGEs, highlighting their role in AMR dissemination.

periods suggests that not only are MGEs responsible for the dissemination of resistance, but they also play a dynamic role in the amplification of ARGs during critical time frames of increased antibiotic use or infection pressure.

3.5. Resistance mechanism

The antimicrobial resistance genes (ARGs) identified in our analysis primarily employed five key resistance mechanisms: *antibiotic efflux* (49 %), *antibiotic inactivation* (15.1 %), *antibiotic target alteration* (14.6 %), *antibiotic target protection* (7.8 %), and *antibiotic target replacement* (4.09 %). These mechanisms represent the predominant strategies through which bacterial populations resist the action of antibiotics, with efflux being the most common, enabling bacteria to pump out antibiotics before they can exert their effects (Poole, 2007). In addition to these primary mechanisms, we also detected a range of combinatory and less common resistance mechanisms. These included *antibiotic efflux and target alteration* (0.20 %), *antibiotic efflux with reduced permeability to antibiotics* (0.06 %), *antibiotic target alteration and replacement* (6.42 %), *reduced permeability to antibiotics* (0.40 %), *reduced permeability combined with resistance by absence of target* (0.10 %), *resistance by absence of target* (0.92 %), and *resistance by host-dependent nutrient acquisition* (1.09 %) (Fig. 5). The prevalence of antibiotic efflux mechanisms aligns with global patterns, where efflux pumps are increasingly recognized as a primary defense against a wide range of antibiotic classes (Sun et al., 2014). However, the significant presence of *target alteration and replacement* mechanisms suggests that bacteria are also evolving their internal structures to avoid antibiotic action, making these genes particularly concerning in the context of multidrug resistance. Overall, our results provide a comprehensive view of the diverse resistance strategies employed by bacteria in wastewater environments,

emphasizing the importance of continuous monitoring to track the evolution and spread of ARGs across different resistance pathways.

4. Conclusion

This study demonstrates the utility of wastewater-based surveillance for monitoring microbial diversity and antimicrobial resistance (AMR) dynamics, particularly in response to the COVID-19 pandemic. Our findings show stable microbial communities with significant shifts during COVID-19 waves, highlighting interactions between viral and bacterial populations. Temporal fluctuations in AMR gene abundance indicate the influence of antibiotic use and seasonal patterns on resistance. Notably, resistance genes against multidrug, macrolide-lincosamide-streptogramin, beta-lactams, and tetracyclines were prevalent, especially in WHO priority pathogens like Enterobacteriaceae and Pseudomonas. Wastewater monitoring offers a valuable tool for tracking AMR trends and pathogen dynamics over time. This study emphasizes the need for integrated surveillance to better understand the environmental spread of AMR and to inform public health strategies. Continued research in this area is essential to address the growing AMR threat in urban settings.

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CRediT authorship contribution statement

Vinay Rajput: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation. **Rinka Pramanik:** Writing – review & editing, Methodology. **Kiran Nannaware:** Writing – review & editing, Methodology. **Priyanki Shah:** Project administration, Funding acquisition. **Asim Bhalerao:** Methodology. **Nidhi Jain:**

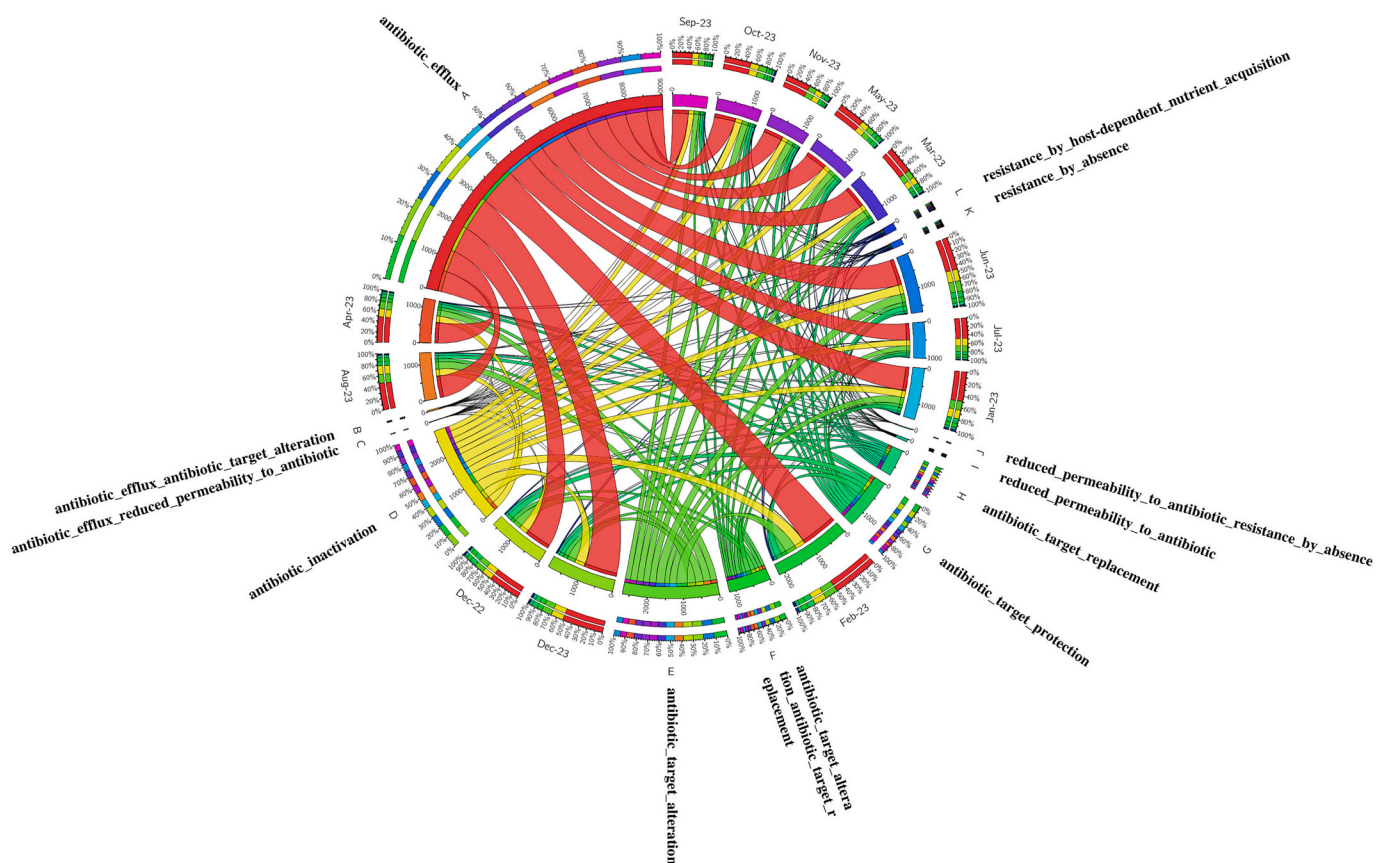


Fig. 5. Abundance of antimicrobial resistance mechanisms in community wastewater. Graph showing the relative and absolute abundance of AMR mechanisms, including efflux pumps, target modification, and enzymatic inactivation, contributing to antimicrobial resistance.

Methodology. L.S. Shashidhara: Supervision, Project administration, Investigation, Funding acquisition. **Sanjay Kamble:** Resources. **Syed Dastager:** Resources. **Mahesh Dharne:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Ethical approval

Ethical approval was not required since study was related to community-based wastewater surveillance in environmental samples.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors confirm that the data supporting the findings of this study are available within the article and its additional information. All raw sequencing data will be available via the NCBI Sequence Read Archive under the Bio Project ID PRJNA1214531.

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